

Chandan Kumar

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EDUCATION

Motilal Nehru National Institute of Technology Allahabad

2021 – 2025

B.Tech. in Computer Science and Engineering

CPI: 8.23

- **Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Distributed Systems, OOP

St. John Senior Secondary School, Dumraon

2018 – 2020

Class XII, CBSE

92.2%

EXPERIENCE

Marvell Technology

May 2024 – Jun 2024 (Intern) | Jan 2025 – Present (Full-Time)

Software Engineer

Pune, India

- Developed Python-based automation frameworks and reusable tooling using PyTest and Paramiko for validating NVMe storage systems, reducing manual effort by approximately 90–95%.
- Built reusable Python scripts and internal utilities to automate firmware flashing, remote VM orchestration over SSH, namespace/VF provisioning, and bring-up workflows, reducing repetitive manual effort and improving engineering productivity.
- Leveraged Docker on Ubuntu to orchestrate parallel storage simulation environments, enabling concurrent execution of independent validation workloads.
- Collaborated with firmware, systems, and cloud engineering teams to debug complex storage issues and ensure reliability across NVMe and virtualization components.
- Integrated automation into CI/CD pipelines, enabling continuous regression testing and early detection of software and firmware issues.
- Performed root-cause analysis using log inspection and system-level debugging to identify failures and improve platform stability.

PROJECTS

Video Summarization | *AI-powered video summarizer*

Nov 2024 | [GitHub](#)

- Developed an AI-powered video summarization pipeline using BiLSTM/LSTM models with attention mechanisms and evaluated it on the SumMe and TVSum benchmark datasets.
- Used Particle Swarm Optimization to improve which parts of a video get picked for the summary.
- **Stack:** Python, PyTorch, LSTM/BiLSTM, OpenCV, ResNet, NumPy, Pandas.

Complaint Box | *Daily life problem solving website*

Nov 2022 | [Live](#) | [GitHub](#)

- Developed a full-stack complaint management platform with secure authentication, email verification, password reset, and role-based access control for users and administrators.
- Designed RESTful APIs and implemented secure authentication with email verification and role-based access control using Node.js and MongoDB.
- **Stack:** Node.js, MongoDB, JavaScript, EJS, CSS, HTML.

SKILLS

Languages : C++, Python

Backend : Node.js, Express.js, REST APIs, PyTest, Paramiko

Systems : Linux (Ubuntu), Bash scripting, Docker, Git, Gerrit, Jira, Confluence, CI/CD pipelines

Core CS : Data Structures, Algorithms, OOP, Operating Systems, DBMS, Distributed Systems

Databases : SQL, MongoDB

ML / AI : PyTorch, LSTM/BiLSTM, OpenCV

ACHIEVEMENTS

LeetCode Knight (**Top 5%**, Rating 1871) | CodeChef Starters 88 — Global Rank **468**